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Poverty, labor markets and trade liberalization in Indonesia

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ABSTRACT

We measure the effects of trade liberalization over the period of 1993–2002 on regional poverty levels in 259 Indonesian districts, and investigate the labor market mechanisms behind these effects. The identification strategy relies on combining information on initial regional labor and product market structure with the exogenous tariff reduction schedule over four three-year periods. We add to the literature on local labor market effects of trade policies by distinguishing between tariffs for output markets and for intermediate inputs, and finding that poverty reduced especially in districts with a greater sector exposure to input tariff liberalization. Among the potential channels behind this effect, we show that low-skilled work participation and middle-skilled wages were more responsive to reductions in import tariffs on intermediate goods than to reductions in import tariffs on final outputs. These results point towards increasing firm competitiveness as a driving factor behind the beneficial poverty effects.

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1. Introduction

Trade liberalization has been widely expected to contribute substantially to poverty reduction in developing countries (e.g., the Doha Ministerial Declaration, WTO, 2001). Under a more open trade regime, rising demand for unskilled labor could benefit poor workers by increasing workers' real wages (Stolper and Samuelson, 1941) as well as creating more jobs in the formal economy. However, the growing body of micro-empirical evidence on the welfare implications of trade liberalization is not unequivocal.¹ Short- to medium-run labor market effects of liberalized trade seem to be very much context specific and depend, among other things, on the previous structure of protection (Attanasio et al., 2004), regional market access (Chiquiar, 2008) as well as the degree of market flexibility. For example, overregulated local labor markets that inhibited the adjustment to structural change could explain the unfavorable regional poverty effects of trade reform in India (Topalova, 2010).² By contrast,

bilateral trade liberalization between the US and Vietnam led to clear reductions in Vietnamese rural poverty, potentially also due to higher labor market mobility (McCaig, 2011). In this latter case, poverty reduction resulted from large improvements in the access to the US market whereas the loss of import protection to local markets was negligible.

Studies focusing on labor market and wage effects of tariff reductions present indirect evidence on potential effects of trade liberalization on poverty, again with mixed results. Reductions in protection and increased foreign competition generally seem to have increased skill premia in Latin America (e.g., Attanasio et al., 2004; Galiani and Sanguinetti, 2003; Goldberg and Pavcnik, 2005), although with some exceptions (e.g., Gonzaga et al., 2006 for Brazil). While most of these studies focus on formal manufacturing employment, Goldberg and Pavcnik (2003) also document an increase in informality in the sectors most exposed to tariff cuts in Colombia. Gaddis and Pieters (2014) find negative effects of trade liberalization for low skill employment in Brazil, and re-allocation of high skill labor from the tradable to nontradable sectors. These empirical findings of increases in skill premia and informality in Latin America suggest that it is less likely that trade would have had strongly favorable poverty effects in the region. However, contrasting evidence is presented by Porto (2006) who finds pro-poor distributional effects of Mercosur in Argentina through price changes and wage responses.

Indonesia offers an interesting case to study the poverty effects of trade liberalization. It is considerably more abundant in unskilled labor than large Latin American countries such as Mexico or Brazil and

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¹ See e.g., Goldberg and Pavcnik (2007) and Winters et al. (2004) for surveys of the earlier literature.

² In a similar vein, tariff reductions in Brazil were associated with increases in urban poverty, which anecdotal evidence attributes to adjustment frictions and rising urban unemployment (Castilho et al., 2012).

hence has a more pronounced comparative advantage in unskilled-labor intensive goods. In the period that we will study in this paper, Indonesia also had relatively flexible labor markets that could potentially restrict the adverse effects of trade reforms on poverty. Moreover, its vast geographic and economic diversity yields potentially large regional variation in the effects of trade liberalization.

With the completion of the Uruguay round in 1994, Indonesia committed itself to substantially lower its remaining tariff barriers across all tradable goods over the following ten years. The tariff reductions were concentrated in the hitherto most protected sectors and resulted in an overall convergence of sectoral protection levels; average import tariff lines decreased from around 17.2% in 1993 to 6.6% in 2002 (see Fig. 1). During the same period, poverty rates also declined, although it is a priori unclear to what extent this decrease can be attributed to trade liberalization.

The existing empirical evidence suggests that trade liberalization could potentially explain a part of the reductions in Indonesian poverty during the nineties. Amiti and Cameron (2012) show that industrial skill premia (defined as the relative wage bill of nonproduction to production workers in manufacturing establishments with at least twenty employees) decreased as a response to tariff reductions. By distinguishing between tariffs on output and intermediate goods used by those firms they are also able to show that skill premia changed mostly because of improved firm competitiveness due to reductions in tariffs on intermediate goods. As a consequence, we would therefore also expect differential effects of tariff reduction through input and output markets on poverty.

Kis-Katos and Sparrow (2011) document that child labor decreased faster in districts that were relatively more exposed to trade liberalization, with indirect evidence that this was driven by positive income effects for the poor. Descriptive evidence also shows the presence of ongoing structural change and reductions in wage inequality (Suryahadi, 2003) as well as improvements in labor conditions (Robertson et al., 2009) over the same time period. However, this evidence, although suggestive, does not directly address the poverty effects of trade liberalization and the relative importance of the different channels for poverty reduction.

In this study we assess the causal effects of tariff reductions on poverty in Indonesian districts in the period of 1993 to 2002. Our study extends the literature on the poverty effects of trade liberalization by explicitly distinguishing tariffs for output markets and for intermediate inputs, and analyzing the effects of reducing these tariffs in a

geographically diverse Southeast-Asian country with large labor mobility. Using district pseudo-panel data, we find that especially reductions in tariffs on intermediate inputs tend to reduce the extent of poverty.

In addition, our analysis focuses on the channels of labor market dynamics, wages, job creation and displacement. With regard to wage effects and job creation, we investigate the regionally differential effects of tariffs on output and intermediate goods using firm level data and household and labor market surveys. We find that increased competitiveness of firms due to lower import tariffs on intermediate goods is weakly related to increases in manufacturing wages. However, we also find evidence of increased poverty due to reduction in output tariffs, presumably due to costly adjustment to trade. This contributes to the empirical evidence on the effects of trade liberalization on local labor markets, in particular highlighting the differences in the mechanisms of liberalization affecting intermediate and output goods.³

The next section describes the context and trends in tariff reductions and poverty and outlines our expectations with respect to the effects of trade liberalization. Section 3 presents the data sources for the pseudo-panel analysis and Section 4 outlines the identification strategy. The results follow in Section 5, while Section 6 investigates the possible confounding trends, discusses caveats and potentially remaining sources of bias. Section 7 concludes.

2. Trade liberalization in Indonesia and its expected effects

2.1. Descriptive trends

Indonesia started to liberalize its trade regime from the mid-1980s, involving a first reduction in tariff lines and a slow tariffication of non-tariff barriers (Basri and Hill, 1996). These reforms were accompanied by reforms of fiscal policy, tax reforms and financial deregulation. The second wave of trade liberalization started in the beginning of the 1990s. By the end of the Uruguay round, Indonesia entered formal multilateral agreements to apply binding tariff ceilings of maximum 40% on 95% of its products (up from 9% of binding tariff ceilings before) (WTO, 1998).

Fig. 1 shows the reduction in average unweighted effectively applied tariff lines across the 1990s: on average, tariff lines reduced from 17.2% in 1993 to 6.6% in 2002. The tariff reductions were not gradual but occurred more or less in two steps over the analyzed time period: the first large reduction of tariff barriers came about with Indonesia's WTO obligations preceding the formation of the WTO, while a second substantial wave of tariff reductions followed in the post monetary crisis period as part of the IMF conditionality package, starting with 1999. Table 1 shows the detailed evolution of the tariffs for the 20 major tradable sectors, which are defined according to a concordance of tariff information and census labor market data. Fig. 2 plots the average reductions in tariffs for these sectors over the entire time period. The high correlation between initial tariff levels and tariff reductions shows that tariff reductions occurred across the board and were the highest in those industries that started with the highest original tariff levels. Moreover, the pattern of tariff reductions across sectors shows that highly protected sectors were not favored by means of delaying exposure to tariff

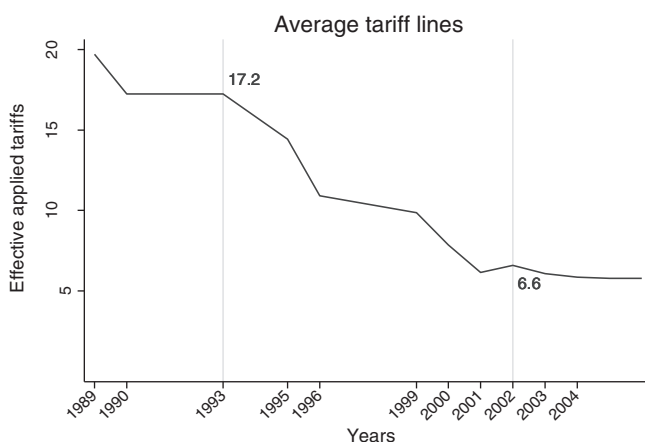


Fig. 1. Evolution of average tariff lines 1993–2002.
Source: Kis-Katos and Sparrow (2011).

³ Goldberg and Pavcnik (2003) assess the effects of trade liberalization on the informal sector in Brazil and Colombia. Autor et al. (2013) look at job displacement in the US due to imports from China, while Iacovone et al. (2013) find displacement effects in Mexico as a result of increased competition from China for its exports on US markets.

Table 1
Evolution of average effectively applied tariff rates by sector.
Source: UNCTAD-TRAINS database.

	1993	1996	1999	2002
<i>Agriculture</i>				
Plants and animals	17.1	12.0	10.3	4.8
Forestry	7.7	3.9	3.5	3.8
Hunting	5.3	4.3	2.2	2.7
Sea fishery	24.9	16.6	14.0	5.2
Fresh-water fishery	10.0	0.0	0.0	0.0
<i>Mining</i>				
Coal mining	5.0	5.0	5.0	5.0
Metal ores mining	3.3	3.2	3.5	2.8
Stones and sand mining	7.0	5.6	3.6	3.5
Salt mining	20.0	15.0	15.0	7.4
Minerals and chemicals mining	2.9	3.0	3.0	2.7
Other mining	4.0	3.4	3.6	3.6
<i>Manufacturing</i>				
Food, beverages, tobacco	23.4	18.1	17.1	12.6
Textiles, apparel, leather	26.0	20.1	16.5	9.4
Wood and products	30.0	16.6	14.1	7.7
Paper and products	20.2	9.5	8.1	4.8
Chemicals and products	11.9	9.4	8.7	6.1
Non-metallic mineral products	20.4	9.5	7.0	5.6
Basic metals	10.3	8.0	7.6	6.4
Metal products	15.8	8.1	7.9	4.9
Other manufacturing	32.0	18.9	18.4	9.6

Note: Sectors are defined based on a concordance between tariff and census labor market data.

reductions.⁴ Due to the firm-specific nature of favoritism and protection in 1990s Indonesia, there were no substantial exemptions from tariff cuts across sectors (Basri and Hill, 1996). In particular, some manufacturing sectors (such as wood, textiles or other manufacturing) with high initial average levels of protection saw average tariff rates reduce to below 10% by 2002.⁵

The period before the economic crisis was also characterized by high labor market flexibility and a highly elastic supply of unskilled labor (Manning, 2000). The early 1990s saw a continued shift from agricultural towards urban employment, an expanding service sector and the growth of an export-oriented economy. These structural changes were accompanied by steadily decreasing poverty rates. Suryahadi et al. (2009) argue that the growth in urban services was a powerful driving force behind these poverty reductions. Increases in inequality during this period suggest, however, that the beneficial effects of the reforms were not concentrated on the very poor (Miranti, 2010). At the same time, labor regulation started to tighten somewhat, with rising minimum wages and extensions of social security coverage.

The 1997/1998 East Asian financial crisis poses a potential challenge to our study as a confounding factor as it overlaps with our second period of analysis (1996–1999), which was also the time period of halting trade liberalization. The crisis had its roots in a monetary contagion leading to a large outflow of foreign capital, currency depreciation, as well as short-term agricultural price hikes. This also led to a sudden increase in expenditure poverty and a temporary restructuring of the labor force towards subsistence production in agriculture. The crisis' impacts were geographically clustered (Java being most strongly hit), but did not differ considerably between rural and urban districts or by the initial levels of poverty (Wetterberg et al., 1999). The extent of expenditure poverty peaked around November of 1998 and declined sharply afterwards, with a quick recovery in consumption growth

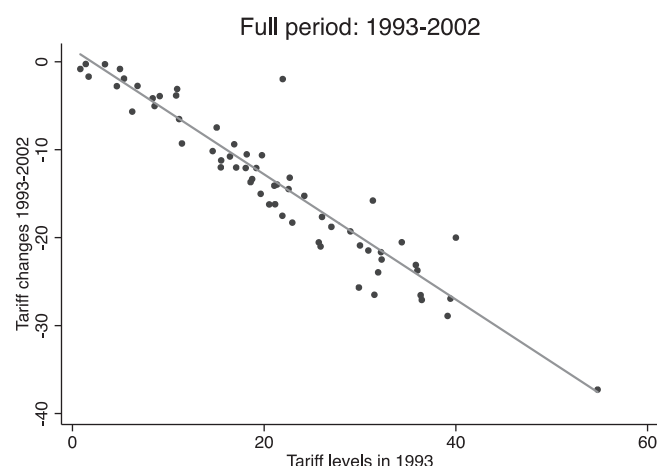


Fig. 2. Tariff reductions by sector 1993–2002.

(Suryahadi et al., 2003). According to our poverty data, the average regional share of those living below the poverty line decreased from 1993 to 1996 from 30 to 25%, jumped to 33% in 1999 and was again at 20% in 2002. We will investigate the importance of the crisis for our results in Section 6.

2.2. Main hypotheses

The primary interest of our study lies in understanding how regional exposure to tariff reductions affected regional levels of poverty in Indonesia, a labor abundant economy. Following the insights of Amity and Cameron (2012) and Amity and Konings (2007), we distinguish between tariffs on output products and tariffs on the intermediate inputs used in local production.

The poverty reducing effects of international trade will be primarily transmitted through labor market mechanisms, in terms of increased wages and employment. According to the neoclassical theory of comparative advantage, under full employment and intersectoral labor mobility, reduction in trading costs can be expected to increase specialization in the production of unskilled labor intensive goods.⁶ This should lead to relative improvements in the wages of the less skilled population and by that to reductions in poverty.

However, labor market adjustments to trade liberalization are most likely to be sluggish and impose a burden on specific groups of workers (Dix-Carneiro, 2014). In the short run, workers are less mobile between sectors and tariff reductions on outputs that put competitive pressures on firms will hurt the workers in the affected sectors disproportionately. In regions where labor is concentrated in those sectors that are losing most of their protection from imports this mechanism will translate to regional wage losses, or to increases in regional unemployment (Hasan et al., 2012). The poverty effects of output tariff reductions depend on the speed of labor market adjustments. In the longer run, a specialization in the unskilled intensive production should reduce poverty. In the short run however, poverty may increase with output trade liberalization if the poor work disproportionately in sectors losing protection. Thus, from a theoretical perspective, we expect output tariff liberalization to reduce regional wages and employment conditional on the speed of the sectoral adjustment of labor. The poverty effects would depend on the extent to which the employment and income of the poor

⁴ Fig. 5 in the Supplementary appendix plots the per-period information presented in Table 1: it shows clearly that in both periods 1993–1996 and 1999–2002 tariff reductions were concentrated in high-tariff sectors and were highly correlated with initial tariff levels.

⁵ The food sector is an exception (with an average tariff of 12.6% in 2002), partly because of tariffication and later exemption of alcoholic beverages.

⁶ This mechanism could be offset if trade protection is concentrated in labor intensive sectors, in which case tariff reduction could reduce the price for unskilled labor intensive goods. However, exploratory regressions show no evidence of previous protection being concentrated in unskilled labor intensive sectors in Indonesia. The results are presented in the Supplementary appendix (Table 14).

is directly affected by the loss of protection but also on the adjustment of incomes in the nontradable sectors.

By contrast, reductions in tariffs and hence prices on inputs that are used in regional production should have the opposite effects on wages, employment and poverty. Lower input tariffs have a more direct productivity enhancing role by rendering inputs cheaper (Amiti and Konings, 2007). Input (but not output) tariff reductions have also been shown to go along with reductions in industrial skill premia (Amiti and Cameron, 2012). As a result, we expect to see short run benefits from productivity gains due to input tariff reductions in form of wage increases and job creation, especially for the workers employed in the sectors affected by input price reductions. How far these benefits translate to overall regional development and reductions in poverty is an empirical question.

We thus decompose the total effect of trade liberalization on poverty into an effect transmitted through a competitive pressure on output product markets and an improving productivity through input trade liberalization.⁷

3. Data

We measure the extent of trade liberalization by reductions in the average (unweighted) tariff lines in 20 tradable goods sectors for the years 1993, 1996, 1999, and 2002.⁸ The source of the tariff information is the UNCTAD-TRAINS database (retrieved through the WITS system of the World Bank). We combine this tariff data with information on the district level labor market structure before the tariff reform, based on the 1990 Indonesian Census.⁹ It provides the main sector of occupation for each individual in the sample at the 2-digit level. In order to combine tariff data with the information on labor market structure, we compute average tariffs at the same level of product aggregation as the available labor market data; we are thus able to distinguish between 5 subsectors in agriculture (plants and animals; forestry; hunting; sea fishery; freshwater fishery), 6 subsectors in mining (coal; metal ore; stones; salt; minerals and chemicals; other mining) and 9 in manufacturing (food, beverage and tobacco; textile, apparel and leather; wood and products; paper and products; chemicals; non-metallic mineral products; basic metals; metallic products; other manufacturing).

Our primary source of household information is the annual national household survey, *Susenas* (*Survei Sosial Ekonomi Nasional*). This repeated cross-section survey is representative at the level of Indonesian districts.¹⁰ Poverty measures are derived from a comparison of monthly per capita household expenditures with province-specific urban/rural poverty lines (based on Suryahadi et al., 2003). Based on this, we calculate three poverty measures (Foster et al., 1984): the poverty headcount ratio (P0, the ratio of people living under the poverty line), the poverty gap (P1, the aggregated income gap of the poor normalized by the total income needed to reach the poverty line), and the squared poverty gap (P2, depicting the depth of poverty, which is defined as the sum of squared individual deviations from the poverty line of those living below the poverty line, normalized by the squared value of the poverty line income).¹¹

We use additional information from *Susenas* on individuals' employment status. All adults (aged 16 years or older) are considered to be working if they report having a permanent job or having worked at least one hour during the week preceding the survey. We also record the primary sector of work for each individual in the sample and distinguish between agriculture, mining, manufacturing and service sectors and use the household surveys to generate further variables used in placebo regressions and further controls (adult education rates, the share of rural population and adult literacy rates).

A second source of individual level information is the annual labor force survey, *Sakernas* (*Survei Angkatan Kerja Nasional*). This allows us to compute real hourly wages (deflated by the provincial consumer price index), which are not available from the household surveys. *Sakernas* data are representative at the level of 23 provinces.

Information on the number of industrial workers and on total industrial wage payments in each district comes from the annual industrial survey SI (*Survei Industri*) that aims to include all Indonesian firms operating with at least 20 employees. Additionally, we use the data on SI firms to describe the regional industrial structure, which enables us to generate alternative regional tariff exposure measures for 60 tradable output sectors.

Following Amiti and Cameron (2012) and Amiti and Konings (2007), we distinguish between the reduction in tariffs on outputs and intermediate inputs by generating a proxy of the regional sectoral input structure based on regional outputs and a national input–output-table. We use the latest national input–output (IO) table from the time period before the reforms, which distinguishes between 161 input and output sectors. The IO-table is based on the economic census (*Sensus Ekonomi*) of 1990, and has been compiled by Statistics Indonesia (BPS). We combine this information with the regional economic structure (based on either sector labor market shares or industrial production) and our tariff variables.

We use these various data sources to build a balanced pseudo-panel of Indonesian districts, which are classified as either rural districts (*kabupaten*) or municipalities (*kota*). Districts are practical units of analysis for assessing the poverty effects of trade liberalization as they are well defined geographic areas and key administrative units in Indonesia that reflect local labor markets. During the period under study, new districts emerged as a result of district splits. We deal with these splits by applying the 1993 district definition frame.¹² Some of the districts had to be dropped from the sample. After excluding peripheral districts with incomplete or missing socio-economic data (all districts in the provinces of Aceh, Maluku and Irian Jaya) as well as East Timor (which gained independence in 1999) we are left with a balanced panel of 259 districts. These 259 districts are organized into 23 provinces and are spread across 5 main island groups (Sumatra, Java, Kalimantan, Sulawesi, and the remaining smaller islands grouped together). Table 2 presents the descriptive statistics on the changes in our main outcome and tariff variables over time; Tables 12 and 13 in the Supplementary appendix report descriptive statistics for all variables included in the analysis.

4. Methods

Following Topalova (2010), several recent studies identify regionally differential effects of trade liberalization by distinguishing between different levels of regional exposure to trade based on the pre-reform labor market structure of the region (Castilho et al., 2012; Fukase, 2013; Kovak, 2013; McCaig, 2011). The advantage of this method is that it

⁷ Trade liberalization may also affect household welfare through consumer prices. We control for this channel by using province-specific poverty lines to calculate poverty measures as well as including island–year interaction terms in the empirical specification.

⁸ Since tariff data is missing for the years 1994, 1997 and 1998, we base our analysis on four equally spaced time periods.

⁹ We use a 1% random sample available for public use through the IPUMS system (Minnesota Population Center, 2011).

¹⁰ For calculating district level variables we use the population weights provided in *Susenas*.

¹¹ We prefer to use these composite poverty measures to statistics on household expenditures at different parts of the income distribution. When normalized by regional poverty lines, poverty measures become directly comparable across the regions whereas we lack comparable information on regional consumption baskets at other points of the income distribution.

¹² District splits followed almost entirely sub-district boundaries within the relevant district, and did not affect borders with neighboring districts. See Fitriani et al. (2005) for a more complete account of this process. Statistics Indonesia maintains a full list of district codes over time. Maps in Figs. 3 and 4 show the boundaries of 440 districts, filled with information on 259 original districts used in the analysis.

Table 2
Descriptive statistics on changes of dependent and tariff variables over time.

	Av. change per period	SD of change per period
<i>Dependent var.</i>		
P0	−0.032	0.129
P1	−0.009	0.037
P2	−0.003	0.015
Share of working adults	−0.014	0.039
Share of working (no education)	−0.018	0.055
Share of working (primary ed.)	−0.012	0.052
Share of working (jun. secondary ed.)	−0.004	0.060
Share of working (min. sen. sec. ed.)	−0.018	0.053
ln total workers	0.029	0.675
ln total real wage bill	0.027	0.826
ln real wage per worker	−0.002	0.529
ln real hourly wage	−0.028	0.179
ln real hourly wage (no education)	0.107	0.349
ln real hourly wage (primary ed.)	0.102	0.288
ln real hourly wage (jun. secondary ed.)	0.040	0.261
ln real hourly wage (min. sen. sec. ed.)	−0.064	0.207
<i>Tariff var.</i>		
Output tariffs (labor weighted)	−4.408	2.069
Input tariffs (labor weighted)	−3.319	2.059
Output tariffs (manuf. prod. w.)	−3.920	3.361
Output tariffs (manuf. prod. w.)	−3.206	2.174

For the total real wage bill, total workers, real wages per worker and manufacturing weighted tariffs $N = 734$, for other variables $N = 777$.

does not only focus on the manufacturing sector or formal employment but measures the effects of trade liberalization at the household level. To define tariff exposure, it uses the labor structure of local residents based on household surveys, irrespectively of the specific place and geographic location of their work, and hence focuses on tariff effects important for local residents. The main poverty measures are derived from household expenditure surveys, which capture the overall extent of regional poverty. We complement this household based information with data from firm and labor market surveys in order to investigate the labor market mechanisms that are behind these poverty effects. For firm and wage outcomes we alternatively define tariff exposure measures that are weighted by the regional firms' output and input structures.

4.1. Measuring tariff exposure at district level

Our empirical strategy applies measures of district tariff exposure that combine variation over time in nationally determined import tariffs with information on the districts' economic structure in the initial pre-reform period. Output tariffs of district k in year t are computed by weighting the actual average import tariffs of each sector s by the sector's relative importance in the local economy, measured in an initial, pre-reform time period:

$$Tariff_{kt} = \sum_{s=1}^S \left(\frac{Q_{sk,t=0}}{Q_{k,t=0}} \times Tariff_{st} \right). \quad (1)$$

For our main poverty analysis, the relative importance of a sector in a district economy is measured by its regional employment share, and hence the weights are given by the relative share of the employment of the output sector s ($Q_{sk,t} = 1990$) in the total labor force of district k ($Q_{k,t} = 1990$), measured before the reform, in 1990.¹³ Fig. 3 maps the geographic variation in reductions of the output tariff measure across the nine years of analysis for all Indonesian districts included in the analysis.

¹³ Our labor shares are calculated based on $S = 20$ different sectors, see Section 2.

In addition to our main labor market based output tariff measure, we also compute an output tariff measure weighted by the district industrial structure, which enables us to identify the effects of tariff changes on the total wage bill, the real wage per worker and employment of local firms. This alternative tariff measure is based on the output structure of formalized firms with at least 20 employees (from SI, the industrial survey). This second output tariff measure weights national tariffs in sector s in year t by the industry's initial share in district k 's industrial output, $Q_{sk,1993}/Q_{k,1993}$, as recorded in the industrial survey in the initial year 1993.¹⁴

For computing the input tariffs, we rely on a national input–output table from 1990 to generate a measure of regional exposure to input tariffs based on the district sectoral structure:

$$InputTariff_{kt} = \sum_{s=1}^S \left(\frac{Q_{sk,t=0}}{Q_{k,t=0}} \times \sum_{j=1}^J \left(\frac{M_{js,1990}}{M_{s,1990}} \times Tariff_{jt} \right) \right). \quad (2)$$

For this, we weight the tariff on each input good j in year t by the initial share of the j -th industry among the inputs of any output sector s , $M_{js,1990}/M_{s,1990}$. We once again aggregate these input tariff measures across all output producing sectors or industries of the district, which are then weighted by the output sector's initial relative regional importance. Since our input–output data does not vary across districts, we have to assume that the national structure of inputs adequately describes the regional input structures, at least on average. By using a pre-reform input–output table we can ensure that tariff induced shifts in the sectoral structure are not reflected in the measure. We capture the relative importance of specific sectors, $Q_{sk,t} = 0/Q_{k,t} = 0$, once again either through labor market shares in 1990 (for household based measures) or shares in output production in 1993 (for firm employment and wages).¹⁵ Fig. 4 maps the regional variation in changes in import tariffs on input goods.

The presence and size of nontradable sectors poses a challenge to the computation of weighted tariff measures. The literature on regional labor market effects of trade liberalization has applied two different approaches to deal with the nontradable sector. In the first, the share of nontradable sectors has been included in the denominator (the size of the local economy) but receives a zero tariff, which would imply no tariff reductions and hence no price changes in nontradable sectors over time (see e.g., McCaig, 2011; Topalova, 2010).¹⁶ Alternatively, excluding the nontradable sectors from the weighting altogether and basing the weights on the sectoral composition of tradable industries assumes that the price adjustments are passed through to the nontradable sectors.

Kovak (2013) shows that under the assumptions of full employment and perfect labor mobility between sectors but not between regions, wages in nontradables fully mirror the wage developments in the tradable sectors. For consistency, nontradables should therefore be excluded from the analysis altogether. Although his theoretical analysis does not have similarly clear predictions for measures of poverty, the argument applies to poverty analysis as well as long as the main poverty effects are transmitted through wage changes.

In our analysis we follow the approach suggested by Kovak (2013). For Eq. (1) this implies that nontradables are excluded from the output tariff variables, since the nontradables do not appear in the variable $Tariff_{st}$. For the input tariff this is more complex. The nontradable sector still appears in Q_s and M_s , to take into account the effects of tariff changes on tradable inputs that are used in the nontradable sector. That is, in

¹⁴ After taking tariff and input–output table concordances into account, we are able to distinguish between $S = 60$ different tradable output industries.

¹⁵ Due to differences in sector definitions and concordances between different sectors, we are able to distinguish between 12 (77) input sectors for the labor (manufacturing output) weighted tariffs.

¹⁶ Topalova (2010) instruments tariffs weighted by labor market shares that include nontradables with tariffs weighted by labor market shares in tradable sectors only.

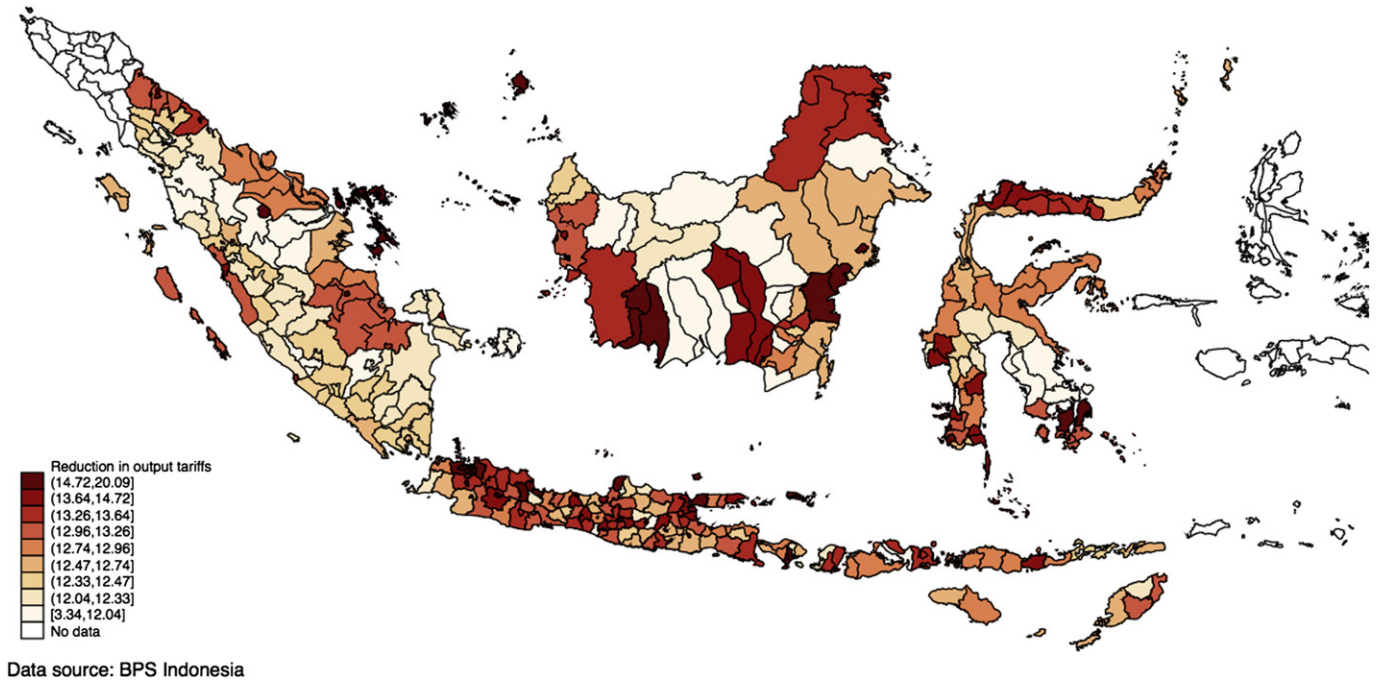


Fig. 3. Reductions in labor weighted district output tariff measures (1993–2002).

Eq. (2) the nontradable sector is dropped from the set J but not from S : we are first weighting the sectoral tariffs by the relative weight of each input among all tradable inputs, but consider both tradables and nontradables among the output sectors. By contrast, the input tariff weighted by manufacturing sector output shares includes tradable goods only since all industrial products included in the weighting scheme are tradable.

4.2. Empirical specification and identification

Our main estimating equation takes the following first difference specification:

$$\Delta y_{kt} = \alpha + \beta_1 \Delta \text{OutputTariff}_{kt} + \beta_2 \Delta \text{InputTariff}_{kt} + \Delta X'_{kt} \gamma + I'_k \theta + \lambda_{rt} + \Delta \varepsilon_{kt},$$

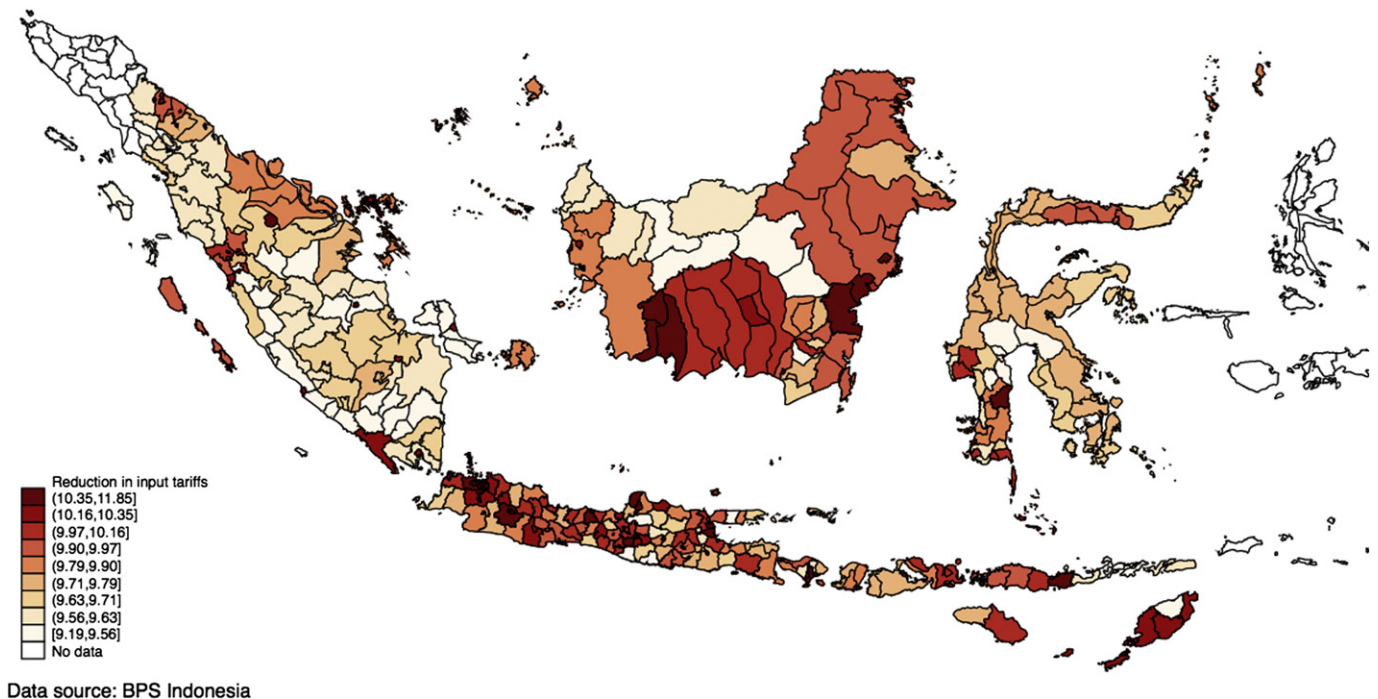


Fig. 4. Reductions in labor weighted district input tariff measures (1993–2002).

where y_{kt} denotes the district level dependent variables (various measures of poverty, employment and wages). X_{kt} is a vector of time variant control variables (share of rural population, share of population aged 16–60, adult (20+) literacy rates, minimum wages). The vector of initial conditions, I_k , includes the 1990 labor shares in the district that are used as tariff weights, aggregated to one digit sectors, the 1990 rural population shares, and, in some specifications, the initial levels of the dependent variable. Time and island interaction terms, λ_{it} , are included to control for island-group specific time effects.¹⁷

The difference specification addresses the potentially endogenous nature of the components in the tariff exposure measure. First, the potential bias due to endogenous tariff setting at the national level is eliminated by controlling for national variation over time and by considering only within-district variation. Second, by taking first differences and removing district fixed effects, we purge any bias due to unobserved heterogeneity that might be introduced by the initial district sectoral structure in employment and industry output. Moreover, the district labor and industry output shares by sector are taken at 1990 and 1993 values respectively and are therefore not directly influenced by district poverty profiles and labor market developments after 1993.

The data captures variation in changes to tariff exposure across districts and within districts across time.¹⁸ Our empirical strategy relies on the identifying assumption that there are no unobserved time variant confounders. This assumption will be violated if poverty trends and labor market dynamics are related to the initial sectoral composition of district economies. The most relevant potential confounding trends include structural change, overall economic development and social policies. Structural change involves a gradual shift from agriculture to manufacturing and service sectors. The extent and speed of such structural change may vary by the initial size of the agricultural sector and the share of the population living in rural areas. Changes in poverty incidence will also be driven by overall economic development as well as targeted social policies. These may vary by initial levels of poverty (due to convergence or policy targeting) and by local economic structure.

We deal with these potential confounding trends by adding time variant controls and controls for initial conditions. The time variant controls capture regional changes in poverty due to shifts in population structure, average skill levels, urbanization and social policies. Some of these controls, for instance urbanization or minimum wages, may however also serve as transmitting factors for the effects of trade liberalization and hence we test whether our results are sensitive to the exclusion of these controls. The controls for initial conditions include initial sectoral labor shares (measured at the one-digit level) as well as the share of rural population in 1990. As an additional sensitivity check we include the 1993 value of the dependent variables (P0, P1 and P2) to proxy for convergence and poverty targeting.

The exogeneity assumption is also violated if the regional variation in trade liberalization is correlated with pre-existing trends. This could be the case if the tariff reductions are influenced by political pressure of industry or local power structures, which would involve specific sectors and districts being only affected by reduction in protection in the later periods. For the 1990s such influence is unlikely to come from the districts directly, as power in Indonesia was heavily centralized. Sector-based lobbying is also less likely as trade policy would be mainly influenced by crony capitalists' rent seeking (Basri, 2001; Basri and Hill, 1996). This is also supported by the strong correlation between tariff cuts and initial tariff levels in the two sub-periods where the bulk of

tariff cuts was concentrated, which suggests that highly protected sectors were not spared from tariff cuts in the early stages of the reforms (see Figs. 2 and 5 in the Supplementary appendix).

To assess potential confounding factors from pre-existing trends we conduct placebo tests of the effects of tariff reductions on past poverty levels, using data from the 1984, 1987 and 1990 *Susenas* household consumption survey. These pre-1993 surveys are limited to consumption data and we can therefore only control for district fixed effects, rural shares and time–island interactions. We regress the changes in poverty rates over the three past periods on future tariff changes between 1993–1996 and 1999–2002, the two sub-periods with the most pronounced variation and strongest trends in tariff reductions.¹⁹ In a similar vein, we also conduct placebo tests for the total nominal wage bill, total workers and nominal wages per worker using three equally spaced rounds of past firm surveys (1983–1986–1989). Finally, we test for confounding trends in current socio-economic outcomes that would not likely be directly affected by trade liberalization in the short term. For this we assess the effect of tariff reductions on education attainment of adults (30–60 years old), which could only be affected by substantial migration but not by schooling.

The 1997/1998 financial crisis poses a potential problem for our empirical strategy since the post-crisis recovery remains a potentially confounding effect. We deal with this concern in two ways: we include in all regressions island–year fixed effects that distinguish between five main geographic units and allow the crisis effects to vary across the main islands. Given the empirical evidence on the strong geographical clustering of the poverty effects of the crisis (Wetterberg et al., 1999), we are able to capture a part of the crisis effects already through this strategy. Robustness checks directly control for the local intensity of the crisis by measuring the differential amplitude of the explosive price increases that were observed during the economic crisis, and have subsequently been associated with the strong poverty increase in 1998 and 1999 (e.g. Hardjono et al., 2010). The crisis variable measures the amplitude of the consumer price index (CPI) across provinces at the height of the crisis (in 1998), which is interacted in the regressions both with the post-crisis year 1999 and with 2002. Additionally, we discuss the sensitivity of our results to estimating the models for separate pre- and post-crisis periods, for the pre- and post-crisis periods jointly, and for the long difference from 1993 to 2002.

5. Results

5.1. Poverty

The general effects of tariff reductions on our three poverty measures (P0, P1 and P2) for different specifications are shown in Table 3. Overall, there is a negative correlation between the three poverty measures and output tariff exposure, implying that tariff reduction for output markets is associated with an increase in poverty. The relationship persists after controlling for year–island interactions, time variant controls, and initial labor force and rural population shares.²⁰ By contrast, input tariffs have a positive association with poverty, which implies that their reduction induces a reduction in poverty. For the poverty headcount this effect grows in magnitude and statistical significance when we add the control variables and initial conditions whereas coefficients are stable throughout specifications for the poverty gap and the depth of poverty.

Controlling for initial levels of the dependent variable in column 4 helps to assess whether initial poverty is associated with differential

¹⁷ Although changes in the two tariff variables are correlated, this does not seem to affect our tariff estimates. They are comparable when estimated jointly or in separate regressions. Once island–year fixed effects are controlled for, the residual correlation between input and output tariff changes amounts to around 0.5.

¹⁸ Figs. 3 and 4 show the across-district variation in the reductions of output and input tariffs, Figs. 6 and 7 (in the Supplementary appendix) show the across-district variation in poverty reduction.

¹⁹ By doing so, we eliminate kinks in the tariff schedule due to stalling tariff reductions between 1996 and 1999 (cf. Fig. 1). Alternatively, regressing past poverty changes on tariff changes over the whole period (as a long difference), or on tariff changes between 1993–1996–1999 yields similar results for the placebo tests.

²⁰ Alternative specifications not shown here, such as leaving out time variant controls from columns (3) to (4) do not affect the results either.

Table 3
Poverty effects of tariff liberalization, labor weighted tariffs.

Source	Household surveys (Susenas panel 1993–2002)			
Model	(1)	(2)	(3)	(4)
<i>Panel A</i>	<i>Dependent: P0</i>			
Output tariffs	–0.0117** (0.0048)	–0.0134** (0.0046)	–0.0122* (0.0050)	–0.0125* (0.0049)
Input tariffs	0.0111 (0.0157)	0.0176 (0.0157)	0.0325* (0.0158)	0.0328* (0.0156)
Test on $OutputTariff + InputTariff = 0$ (p-val.)	0.959	0.756	0.149	0.143
<i>Panel B</i>	<i>Dependent: P1</i>			
Output tariffs	–0.0037** (0.0013)	–0.0040** (0.0013)	–0.0034* (0.0014)	–0.0035* (0.0014)
Input tariffs	0.0093* (0.0047)	0.0108* (0.0048)	0.0164** (0.0048)	0.0167** (0.0047)
Test on $OutputTariff + InputTariff = 0$ (p-val.)	0.163	0.100	0.003	0.002
<i>Panel C</i>	<i>Dependent: P2</i>			
Output tariffs	–0.0014** (0.0005)	–0.0015** (0.0005)	–0.0012* (0.0006)	–0.0013* (0.0005)
Input tariffs	0.0049** (0.0019)	0.0056** (0.0020)	0.0078** (0.0020)	0.0080** (0.0020)
Test on $OutputTariff + InputTariff = 0$ (p-val.)	0.026	0.016	0.000	0.000
N observations	777	777	777	777
N districts	259	259	259	259
Year–island dummies	Yes	Yes	Yes	Yes
Time variant controls	No	Yes	Yes	Yes
Initial labor force and rural shares	No	No	Yes	Yes
Dependent variable 1993	No	No	No	Yes

Note: Each block of the table reports tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs and further controls. Time variant controls include first differences of the share of rural population, the share of working age population (16–60), literacy rates at age 20–99 and minimum wage. Standard errors, clustered at the district level, are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

parallel trends that may confound our estimates. We find no evidence of this as all results are robust to including these variables. Since we prefer not to include lagged levels of the dependent variable in a first difference specification, we omit these in the remainder of the analysis. In what follows, we refer to column 3 as our preferred specification for interpreting the economic magnitudes of the effects.²¹

The association between tariff reductions and poverty changes turns out to be of an economically significant magnitude. The point estimates suggest that a one standard deviation (or 2.06 percentage point) larger reduction in the weighted input tariff measure implies a 6.7 percentage point larger reduction in the poverty headcount (P0), which corresponds to about half of a standard deviation of the decrease in the poverty headcount over a three year period (cf. Table 2). By contrast, a one standard deviation larger decrease in output tariffs is associated with about half a standard deviation larger increase in the poverty headcount (the absolute magnitude of the two coefficients is not statistically significantly different). The difference between the tariff coefficients increases for the poverty gap (P1) and the depth of poverty (P2).²² Column 3 further shows that a one standard deviation reduction in the weighted input tariff measure is associated with a decrease of the poverty gap by about 92% of the standard deviation of the average decrease in the poverty gap over time (panel B), and a one standard deviation larger decrease in the depth of poverty (panel C).

²¹ See Table 16 in the Supplementary appendix for the results on additional controls in these specifications.

²² The sum of the two (negative output and positive input) tariff coefficients is not statistically significantly different from zero for the poverty headcount, but it is smaller than zero for the column 3 specifications of the poverty gap and squared poverty gap (see Table 3). The direct comparison of the two tariff coefficients is meaningful since the two tariff reductions have roughly the same standard deviation (2.06 and 2.07).

Table 4
Poverty by education level of the head of household, labor weighted tariffs.

Source	Household surveys (Susenas 1993–2002)			
Household head education	No educ.	Primary educ.	Jun. second. educ.	Senior sec. or more
	(1)	(2)	(3)	(4)
<i>Panel A</i>	<i>Dependent: P0</i>			
Output tariff	–0.0171* (0.0072)	–0.0051 (0.0061)	–0.0073 (0.0063)	–0.0153** (0.0044)
Input tariff	0.0220 (0.0236)	0.0173 (0.0214)	0.0154 (0.0213)	0.0532** (0.0145)
Mean dependent	–0.032	–0.032	–0.010	–0.004
<i>Panel B</i>	<i>Dependent: P1</i>			
Output tariff	–0.0045* (0.0019)	–0.0023 (0.0016)	–0.0016 (0.0016)	–0.0023** (0.0008)
Input tariff	0.0135† (0.0070)	0.0152* (0.0060)	0.0105* (0.0051)	0.0104** (0.0033)
Mean dependent	–0.009	–0.009	–0.003	–0.001
<i>Panel C</i>	<i>Dependent: P2</i>			
Output tariff	–0.0017 (0.0008)	–0.0009 (0.0006)	–0.0005 (0.0005)	–0.0005† (0.0003)
Input tariff	0.0072* (0.0030)	0.0079** (0.0023)	0.0052** (0.0019)	0.0031* (0.0014)
Mean dependent	–0.003	–0.003	–0.001	–0.0001
N observations	777	777	777	777
Year–island dummies	Yes	Yes	Yes	Yes
Further controls	Yes	Yes	Yes	Yes

Note: Each block of the table reports separate tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs and further controls. Specifications are identical to model 3 of Table 3, further controls including first differences of the share of rural population, share of working age population (16–60), literacy rates at age 20–99, minimum wages, as well as 1990 labor shares and 1993 rural population shares. Standard errors, clustered at the district level, are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

Tariff reductions for intermediate inputs seem to have contributed to alleviating the depth of poverty in Indonesia and have been particularly favorable for the very poor, while reducing output tariffs has somewhat smaller offsetting effects. Based on these results, the beneficial effects of cheaper local input prices seem to be large enough to at least neutralize the poverty-increasing effects of output tariff liberalization.

Table 4 disaggregates the results from our preferred specification (column 3 in Table 3) by skill levels of the household heads. We distinguish between four educational categories: household heads who did not complete primary education, those with at most primary education, those with completed junior secondary education, and those with at least a completed senior secondary education. We see that tariff reductions in intermediate inputs reduce poverty across all education levels: for the low-skilled population these effects are observed mainly at the lower end of the income distribution (through their effects on the depth of poverty), while for the high skill population the effects are more prominent closer to the poverty line. We find a reduction in the poverty headcount only for the high skilled population, presumably because the relatively higher educated poor are likely to be concentrated closer to the poverty line than the low educated poor. As before, we see the opposite effects for the output tariffs, with the patterns almost mirroring those for input tariffs yet at smaller magnitudes. This suggests that the adjustment costs of trade liberalization are borne by both low and high skill households.

5.2. Labor market dynamics

Whereas the share of working adults (measured by working status reported in the household surveys) decreased slightly over the time period, this decrease has been mitigated by decreases in tariffs. We find that changes to output and input tariffs both induce higher market participation, although when estimated jointly, as documented in Table 5, only the decrease in tariffs on intermediate inputs remains statistically

Table 5
Working status of adults, by education level, labor weighted tariffs.

Source	Household surveys (<i>Susenas</i> 1993–2002)				
Dependent	Share of working adults				
Sample (adults)	All	No education	Primary education	Jun. second. education	Senior sec. or higher
	(1)	(2)	(3)	(4)	(5)
Output tariff	−0.0022 (0.0025)	−0.0033 (0.0030)	0.0030 (0.0033)	0.0008 (0.0032)	−0.0018 (0.0036)
Input tariff	−0.0145† (0.0082)	−0.0192† (0.0116)	−0.0395** (0.0110)	−0.0182† (0.0106)	0.0108 (0.0102)
Mean dependent	−0.014	−0.018	−0.012	−0.004	−0.018
N observations	777	777	777	777	777
Year–island dummies	Yes	Yes	Yes	Yes	Yes
Further controls	Yes	Yes	Yes	Yes	Yes

Note: The table reports separate tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs and further controls. Specifications are identical to model 3 of Table 3, further controls including first differences of the share of rural population, share of working age population (16–60), literacy rates at age 20–99, minimum wages, as well as 1990 labor shares and 1993 rural population shares. Standard errors, clustered at the district level, are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

significant. A one standard deviation larger decrease in local input tariffs reduces the average drop in work participation by about 77% of its standard deviation. This effect is most pronounced for those with primary education: for adults with no education the effect equals 72%, for those with completed primary education to 156% of its standard deviation. For the middle education category (junior high school), it again decreases both in size (with 62% of a standard deviation) and statistical significance, while it is not statistically significant for the highest educational category. These findings suggest job creation especially for the educated low skilled due to regional growth triggered by increased competitiveness of local firms.²³

5.3. Firms: wages and workers

Previous firm level analyses document that trade liberalization in Indonesia has improved firm productivity (*Amiti and Konings, 2007*) and increased the relative magnitude of the wage bill paid by manufacturing firms to production as opposed to non-production workers (*Amiti and Cameron, 2012*). These effects were in particular due to decreases in import tariffs on intermediate production goods used by the firms. These findings suggest that direct improvements in the profitability of local firms might also help to explain the observed favorable income effects to the poor of import tariff liberalization. In order to investigate this channel more closely, we extend the analysis to the total wage bill of manufacturing firms in the district (deflated using provincial CPI), the total employment by those firms and the real wages they pay per worker. We use the same manufacturing firm data as the two studies above and also differentiate between the effects of tariffs for intermediate inputs and production outputs, but run the analysis at the level of the regional economies in order to retain comparability with our previous results.

Table 6 shows sizeable negative tariff coefficients on firm wages and employment that are imprecisely estimated. As before, the point estimates on input tariff reductions are of a consistently more sizeable magnitude, although the coefficients are statistically not significantly different from zero or each other.²⁴ Nonetheless, together with the findings by *Amiti and Cameron (2012)*, we cannot exclude that productivity

Table 6
Firm wages and employment, tariffs weighted by manufacturing output.

	Source: Firm survey (<i>SI</i> 1993–2002)		
Dependent	In total real wage bill	In total workers	In real wage per worker
	(2)	(3)	(4)
Output tariff	−0.0121 (0.0158)	−0.0028 (0.0133)	−0.0092 (0.0096)
Input tariff	−0.0476 (0.0325)	−0.0173 (0.0257)	−0.0302 (0.0231)
N observations	734	734	734
Year–island dummies	Yes	Yes	Yes
Further controls	Yes	Yes	Yes

Note: Each block of the table reports separate tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs, island–year dummies and further controls. Specifications are identical to model 3 of Table 3, further controls including first differences of the share of rural population, share of working age population (16–60), literacy rates at age 20–99, minimum wages, as well as 1990 labor shares and 1993 rural population shares. Standard errors, clustered at the district level are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

improvements due to trade liberalization have led to wage increases (and potentially job creation) in the formal manufacturing sector. These developments may have reduced poverty of workers directly, but may also have increased regional wage levels for the workers outside the formal manufacturing sector and in the nontradable sectors, reducing poverty further.

Since the *SI* data does not provide information on the hours worked, we cannot clearly distinguish between wages and job creation. However, we can look at average wages at province level, which are collected by the national labor force survey (*Sakernas*). As the number of provinces is considerably lower than the number of districts (23 as compared to 259), this is admittedly a much cruder measure, but can be disaggregated by education level. For each district, we therefore impute the average wage at the province level and relate this to the district level tariff measures. We address the intra-province correlation due to imputing province level average wages for each district by clustering standard errors in these regressions at the province level. As a robustness check we replicate these regressions with the wage variables expressed at the province level, although this reduces the number of observations to 69. In this second set of regressions we only include the year–island dummies as further controls.

The estimates are shown in panel A of Table 7 for the district level analysis (with imputed province average wages) and in panel B for the province level. Both sets of regressions confirm that average wages have increased as a result of tariff reductions for intermediate inputs, while no significant effects could be observed from changes in output tariffs. Interestingly, we detect some asymmetric effects for the different skill categories: it seems that mainly middle skilled workers have benefited from the input tariff reductions while we find no or even reverse wage effects among the low- and high skilled. The estimates in panel B are imprecise and potentially suffer from a small number of provinces.

6. Sensitivity analysis and caveats

6.1. Robustness of poverty results to confounding trends

Three tests for confounding trends are presented in Table 8. Panel A shows the regressions of current adult education rates measured by the share of those who did not complete primary education, completed primary education, junior secondary or at least senior secondary education. We expect these outcomes to be affected mainly by differential migration patterns. The regressions show that input tariff reductions have led to an increase in the share of primary educated, together with a decrease in the share of higher educated. Such differential migration patterns could result from increased demand for lower-skilled

²³ We do not find a statistically significant relationship between changes in unemployment and tariff changes at any skill level and hence the positive effect on the share of working adults is directly mirrored in an increase in adult labor force participation.

²⁴ Estimated on a somewhat larger sample (using specification 2 of Table 3, including 266 instead of 249 districts), the negative input tariff coefficients turn statistically significant at the 10% level both for the total wage bill and for real wages per worker.

Table 7

Hourly wages by education level of workers, tariffs weighted by manufacturing output.

Panel A	Source: Labor market surveys (<i>Sakernas</i> 1993–2002, per district)				
Dependent	ln real hourly wage (provincial averages imputed for districts)				
Sample (workers)	All	No education	Primary education	Junior sec. education	Senior sec. or higher
	(1)	(2)	(3)	(4)	(5)
Output tariff	0.0016 (0.0025)	−0.0014 (0.0031)	−0.0017 (0.0023)	−0.0002 (0.0024)	0.0025 (0.0032)
Input tariff	−0.0089† (0.0046)	0.0086 (0.0078)	0.0041 (0.0045)	−0.0115* (0.0054)	−0.0064 (0.0056)
N observations	777	777	777	777	777
Year–island dummies	Yes	Yes	Yes	Yes	Yes
Further controls	Yes	Yes	Yes	Yes	Yes
Panel B	Source: Labor market surveys (<i>Sakernas</i> 1993–2002, per province)				
Dependent	ln real hourly wage				
Sample (workers)	All	No education	Primary education	Junior sec. education	Senior sec. or higher
	(1)	(2)	(3)	(4)	(5)
Output tariff	−0.0072 (0.0181)	0.0014 (0.0391)	−0.0248 (0.0373)	−0.0016 (0.0152)	−0.0101 (0.0270)
Input tariff	−0.0714 (0.0500)	0.0748 (0.0881)	0.0568 (0.0515)	−0.1561* (0.0655)	−0.0562 (0.0798)
N observations	69	69	69	69	69
Year–island dummies	Yes	Yes	Yes	Yes	Yes
Further controls	No	No	No	No	No

Note: Each block of the table reports separate tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs and island–year dummies. Specifications in panel A are identical to model 3 of Table 3, further controls including first differences of the share of rural population, share of working age population (16–60), literacy rates at age 20–99, minimum wages, as well as 1990 labor shares and 1993 rural population shares. In panel B only year–island dummies are included. Standard errors, clustered at the province level are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

labor by firms that were facing cheaper input prices. However, these migration patterns could not explain the poverty reducing effects that we see associated with tariff reductions: decreases in relatively higher skilled (and less poor) population would lead to an underestimation of the beneficial poverty effects.

The placebo tests in panel B of Table 8 regress changes in past poverty rates (from 1984 to 1987 and from 1987 to 1990) on changes in tariffs over the two major current periods of tariff reform (1993 to 1996 and 1999 to 2002).²⁵ The results show no evidence of any correlation of the tariff decreases with the pre-existing trends in poverty. Note that the regressions control for island–year interactions and the rural population share, but not initial conditions. This suggests that any potential confounding effects are eliminated in our first difference approach.

The placebo regressions for the total nominal wage bill, total workers and nominal wages per worker from three rounds of past firm surveys (1983–1986–1989) also show no statistically significant effects (panel C of Table 8). As before, changes in firm outcomes are regressed on changes in tariffs over two periods. Moreover, whenever the coefficients are somewhat more sizable (input tariffs for the wage bill and total number of workers) they are of the opposite sign than what we would expect if confounding trends were driving our results. Further results on the local firms' internationalization (reported in Table 17 in the Supplementary appendix) show that during the 1990s the share of foreign investment flows in total firm investment and the share of imported input use increased significantly with input tariff reductions. By contrast, increases in foreign investment and imported input shares over the previous decade (1983–1986–1989) were not related to future tariff reductions. This gives us further evidence that our analysis is not affected by confounding pre-trends from previous changes in trade policies.

6.2. The monetary crisis and differential effects over time

The Southeast Asian monetary crisis of 1997/1998 constitutes a potentially important confounding factor during the analyzed time period, especially since it induced a short-term spike in relative food prices and sharp short-term increases in poverty. Since the effects of the crisis were strongly geographically clustered, the inclusion of island–year fixed effects deals partly with this problem. However, in order to exclude that the crisis confounds our estimates, we repeat our main specifications for various sub-periods before and after the crisis. In an alternative approach, we include additional controls that proxy for the geographic variation of the crisis effects. Moreover, we present long-difference estimates over the total time period of 1993 to 2002.

Table 9 shows our preferred specification for the poverty measures in column 1. Column 2 re-estimates the main specification but focuses only on the two time periods of considerable tariff changes and exogenously imposed tariff reductions: 1993–1996 and 1999–2002, excluding the crisis-period 1996–1999. Although poverty reductions in the last time period could be still affected by the post-crisis recovery, this specification eliminates the time period with little change in tariffs and concurrent poverty increases and hence serves as a useful robustness check with respect to the influence of the crisis period. Column 2 shows very stable input tariff coefficients whereas output tariff coefficients reduce somewhat in size, losing statistical significance.

Separate estimation of the tariff effects for the first and third period in columns 3 and 4 reduces precision even further. Output tariff coefficients stay relatively stable for the first time period, but no effects are identified for the third period. These results together with those in column 2 suggest that the variation in our poverty analysis comes from the differential reductions in tariffs across the two sub-periods of trade liberalization. This also explains the weak results in the long difference specification in column 5. Given the results of the placebo and robustness tests, we can only explain this by the fact that a large part of the identifying variation comes from the differential tariff changes

²⁵ Alternatively, the tests of pre-trends also appear not to be statistically significant if we estimate each period separately.

Table 8
Confounding trends and placebo regressions based on household and firm surveys.

Panel A				
Changes in shares of adults per education category (30–60 years old)				
Source	Susenas 1993–96–99–02			
Dependent	No education	Primary ed.	Jun. second. ed.	Sen. sec./higher
	(1)	(2)	(3)	(4)
Output tariff	–0.0005 (0.0027)	0.0054 (0.0035)	–0.0003 (0.0015)	–0.0046† (0.0026)
Input tariff	0.0032 (0.0081)	–0.0218** (0.0083)	–0.0032 (0.0047)	0.0219** (0.0085)
Mean dependent	–0.060	0.016	0.013	0.031
N	777	777	777	777
Year–island dummies	Yes	Yes	Yes	Yes
Further controls	Yes	Yes	Yes	Yes
Panel B				
Placebo: Past changes in poverty rates				
Source	Susenas 1984–87–90			
Dependent	P0	P1	P2	
	(1)	(2)	(3)	
Output tariff	0.0079 (0.0101)	0.0033 (0.0027)	0.0013 (0.0011)	
Input tariff	0.0281 (0.0452)	–0.0026 (0.0159)	–0.0043 (0.0068)	
Mean dependent	–0.009	–0.008	–0.005	
N	294	294	294	
Year–island dummies	Yes	Yes	Yes	
Further controls	Yes	Yes	Yes	
Panel C				
Placebo: Changes in past employment and nominal wages				
Source	SI 1983–86–89			
Dependent	In total wage bill	In total workers	In wage per worker	
	(1)	(2)	(3)	
Output tariff	–0.0103 (0.0263)	–0.0191 (0.0207)	0.0088 (0.0113)	
Input tariff	0.0501 (0.0493)	0.0445 (0.0464)	0.0056 (0.0200)	
Mean dependent	0.722	0.409	0.313	
N	394	394	394	
Year–island dummies	Yes	Yes	Yes	
Further controls	No	No	No	

Note: The table reports separate tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs and further controls. Panel A is based on 4 rounds of contemporaneous household data, panel B is based on 3 earlier rounds of household data, and panel C on 3 earlier rounds of firm data. In panel A further controls include changes in the share of the rural population, the share of working age population (16–60), and minimum wage. Panels B and C regress changes in past outcomes on tariff changes between the years 1993–1996 and 1999–2002 and island–year effects; further controls in panel B include changes in rural share. Standard errors, clustered at the district level, are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

in the sub-periods.²⁶ Thus, we cannot identify the effects of trade liberalization on the regional differences in poverty reduction over the whole time period, although differential reductions in poverty over time seem to be related to trade policy reform.

For the wage and employment effects, eliminating the middle sub-period increases the precision of the firm based estimates, while the labor survey based hourly wage results stay stable (column 2 of

²⁶ This is supported by descriptive statistics in Table 15 of the Supplementary appendix. While the changes in district tariff exposure are substantial (on average 9.96 to 13.33 percentage points, depending on the choice of tariff variable), the variation across districts is relatively small compared to changes for the full panel or the sub-districts. In other words, the within-island variation in the change from 1993 to 2002 has been fairly similar for all districts, which provides little identifying variation for the long differences analysis. However, the path to these overall changes has been different for districts, providing more identifying variation in tariff exposure changes across districts, especially in the sub-periods 1993–1996 and 1999–2002.

Table 9
Poverty effects by time period, tariffs weighted by labor shares.

Source	Household surveys (Susenas)				
Time period	1993–2002 panel	1993–96, 1999–02 diff.	1993–96 diff.	1999–02 diff.	1993–02 long diff.
	(1)	(2)	(3)	(4)	(5)
Panel A <i>Dependent: P0</i>					
Output tariff	–0.0122* (0.0050)	–0.0081 (0.0053)	–0.0124* (0.0061)	–0.0013 (0.0108)	–0.0003 (0.0043)
Input tariff	0.0325* (0.0158)	0.0312† (0.0166)	0.0419 (0.0613)	0.0275 (0.0680)	0.0111 (0.0308)
Panel B <i>Dependent: P1</i>					
Output tariff	–0.0034* (0.0014)	–0.0012 (0.0015)	–0.0026* (0.0017)	0.0025 (0.0034)	0.0006 (0.0012)
Input tariff	0.0164** (0.0048)	0.0159** (0.0048)	0.0009 (0.0203)	0.0103 (0.0245)	–0.0006 (0.0124)
Panel C <i>Dependent: P2</i>					
Output tariff	–0.0012* (0.0006)	–0.0003 (0.0006)	–0.0009 (0.0007)	0.0015 (0.0014)	0.0003 (0.0005)
Input tariff	0.0078** (0.0020)	0.0075** (0.0020)	0.0003 (0.0091)	0.0038 (0.0112)	–0.0006 (0.0056)
(Year–)island dummies	Yes	Yes	Yes	Yes	Yes
Further controls	Yes	Yes	Yes	Yes	Yes
N	777	518	259	259	259

Note: Each block of the table reports separate tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs and further controls. Specifications are identical to model 3 of Table 3, further controls including first differences of the share of rural population, share of working age population (16–60), literacy rates at age 20–99, minimum wages, as well as 1990 labor shares and 1993 rural population shares. Standard errors, clustered at the district level are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

Table 10). Now both the firm wage bill and firm wages per worker show a statistically significant negative correlation with input tariffs, pointing to productivity gains leading to wage increases. The by-period and long-distance estimates are considerably more stable across sub-periods than the poverty results, although they are still not statistically significant throughout all sub-periods.

The results of Table 9 do not yet address the concern that the differences in the post-crisis poverty reductions after 1999 depend on the within-island variation of the severity of crisis effects (not controlled for by island–year effects), which may confound our trade policy estimates. We investigate this issue by including the amplitude of the hike in consumer prices at the height of the crisis in 1998 as a further control, measured by the increase in the consumer price index from 1996 to 1998 in 23 provincial capitals. The crisis proxy is included into the regressions in the form of two crisis–year interaction terms in order to allow both the 1999 increase in poverty and the 2002 recovery to vary separately with the regional severity of the crisis. The results in Table 11 show no evidence of the poverty, wage and employment estimates being affected by differential crisis impacts.²⁷ The robustness of our results to the inclusion of crisis proxies strengthens the argument that we indeed capture the effects of trade policies and not that of confounding economic shocks.

7. Conclusion

We have examined the effects of trade liberalization in Indonesia from 1993 to 2002 on poverty levels in 259 Indonesian districts and

²⁷ The detailed results show that the higher crisis amplitude was more strongly related to slower post-1999 reductions in poverty, although not so much to the variation in poverty increases from 1996 to 1999 (and hence there may be less residual variation left in the crisis effects within islands that is not picked up by our island–year effects). At the same time, crisis amplitude is related to a reduction of employment by firms and an increase of wages per worker in the second period.

Table 10
Employment and wages by time period, tariffs weighted by manufacturing output.

Time period	1993–2002 panel	1993–96, 1999–02 diff.	1993–96 diff.	1999–02 diff.	1993–02 long diff.
Time period	(1)	(2)	(3)	(4)	(5)
Panel A Source: Firm survey (district level)					
Dependent	In total real wage bill				
Output tariff	–0.0121 (0.0158)	–0.0036 (0.0154)	–0.0047 (0.0162)	0.0042 (0.0271)	0.0032 (0.0142)
Input tariff	–0.0476 (0.0325)	–0.0789* (0.0351)	–0.0452 (0.0359)	–0.1441* (0.0630)	–0.0597 (0.0463)
N	734	487	247	240	239
Panel B Source: Firm survey (district level)					
Dependent	In total workers				
Output tariff	–0.0028 (0.0133)	–0.0006 (0.0248)	–0.0038 (0.0139)	0.0095 (0.0244)	0.0064 (0.0102)
Input tariff	–0.0173 (0.0257)	–0.0119 (0.0432)	–0.0229 (0.0270)	–0.0602 (0.0601)	–0.0089 (0.0378)
N	734	487	247	240	239
Panel C Source: Firm survey (district level)					
Dependent	In real wage per worker				
Output tariff	–0.0092 (0.0096)	–0.0050 (0.0101)	–0.0009 (0.0131)	–0.0054 (0.0163)	–0.0032 (0.0087)
Input tariff	–0.0302 (0.0231)	–0.0428† (0.0252)	–0.0223 (0.0299)	–0.0839* (0.0417)	–0.0508* (0.0252)
N	734	734	247	240	239
Panel D Labor market survey (district level, province averages imputed)					
Dependent	In real hourly wage				
Output tariff	0.0016 (0.0025)	0.0025 (0.0025)	0.0011 (0.0038)	0.0037 (0.0033)	0.0004 (0.0027)
Input tariff	–0.0089† (0.0046)	–0.0094† (0.0050)	–0.0130* (0.0065)	–0.0055 (0.0093)	–0.0127 (0.0083)
N	777	777	259	259	259
(Year–)island dummies	Yes	Yes	Yes	Yes	Yes
Further controls	Yes	Yes	Yes	Yes	Yes

Note: Each block of the table reports separate tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs and further controls. Specifications are identical to model 3 of Table 3, further controls including first differences of the share of rural population, share of working age population (16–60), literacy rates at age 20–99, minimum wages, as well as 1990 labor shares and 1993 rural population shares. Standard errors, clustered at the district level (at the province level in panel D) are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

the role of labor market as a channel for these effects. During this period, Indonesia reduced its tariff barriers across all tradable sectors, with average import tariffs decreasing from 17.2% in 1993 to 6.6% in 2002. This period also saw overall reductions in poverty, despite a temporary setback from the 1997/1998 economic crisis.

The identification strategy relies on combining information on initial regional labor and product market structure with the exogenous tariff reduction schedule over three three-year intervals. The results are robust to specification and controlling for initial conditions in labor market structure. The robustness checks and placebo regressions show no evidence of confounding trends, lending support to our identification strategy.

Our results suggest that input trade liberalization has contributed partially to poverty reduction in Indonesia by increasing incomes of the poorest segment of the population. The effects of tariff liberalization and increased competition in the regional output markets tended to increase poverty, while at the same time, tariff reductions for inputs lead to poverty reductions. This highlights the fact that for a short-run analysis of poverty effects of trade liberalization the effects propagated through input markets could also play a relevant role.

Table 11
Controlling for the differential price shock during the 1997/98 crisis.

Panel A		Household surveys (Susenas 1993–2002)				
		(1)	(2)	(3)		
Dependent		P0	P1	P2		
Output tariff		−0.0120* (0.0050)	−0.0033* (0.0014)	−0.0012* (0.0006)		
Input tariff		0.0285† (0.0159)	0.0145** (0.0048)	0.0069** (0.0019)		
N		777	777	777		
Panel B		Firm surveys (SI 1993–2002)				
Dependent		In total real wage bill	In total workers	In real wage per worker		
Output tariff		−0.0125 (0.0158)	−0.0037 (0.0132)	−0.0088 (0.0096)		
Input tariff		−0.0462 (0.0328)	−0.0133 (0.0260)	−0.0329 (0.0230)		
N		734	734	734		
Year–island dummies		Yes	Yes	Yes		
Further controls		Yes	Yes	Yes		
Interactions of 1999 and 2002 w. crisis proxy		Yes	Yes	Yes		
Panel C		Labor market surveys (Sakernas 1993–2002)				
Dependent		In real hourly wage (provincial averages imputed for districts)				
Sample (workers)		All	No educ.	Prim. ed.	Jun. sec. ed.	Senior sec.+
Output tariff		0.0011 (0.0024)	−0.0007 (0.0029)	−0.0025 (0.0022)	−0.0006 (0.0024)	0.0023 (0.0032)
Input tariff		−0.0071 (0.0047)	0.0051 (0.0077)	0.0078† (0.0046)	−0.0097† (0.0055)	−0.0055 (0.0057)
N		777	777	777	777	777
Year–island dummies		Yes	Yes	Yes	Yes	Yes
Further controls		Yes	Yes	Yes	Yes	Yes
Interactions of 1999 and 2002 w. crisis proxy		Yes	Yes	Yes	Yes	Yes

Note: The table reports separate tariff coefficients, generated by first difference estimates of the reported dependent variables on tariffs and further controls. Specifications are identical to model 3 of Table 3, further controls including first differences of the share of rural population, share of working age population (16–60), literacy rates at age 20–99, minimum wages, as well as 1990 labor shares and 1993 rural population shares. Tariff measures are weighted by labor shares in panel A and by manufacturing production shares in panels B and C. The crisis variable measures the amplitude of the provincial CPI at the height of the crisis (in 1998) and is interacted with a year indicator for 1999 and 2002. Standard errors, clustered at the district level (province level in column 6), are reported in parentheses. **, *, and † mark statistical significance at the 1, 5, and 10% levels.

The driving mechanism behind these effects seems to be increasing firm competitiveness as a direct result of reductions in import tariffs on intermediate goods, which induced increased work participation for low and medium skilled labor as well as wage increases for medium skilled labor. These experiences with trade liberalization add caution to the current policy debate in light of the recent surge in protectionist tendencies in Indonesian trade and economic policies (Nehru, 2013).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jdeveco.2015.07.005>.

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